

Wireless and Internet communications technologies for monitoring and control

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Abstract

Over the past 10 years there have been rapid advances in wireless and Internet technologies. In the case of Fieldbuses many years of standardisation effort have failed to produce a single standard. However, the availability of cheap Ethernet-based networking hardware and PC technology driven by the huge office IT market coupled with the explosion of Internet has made Ethernet an attractive proposition for control and monitoring systems. This has led to parallel developments in Industrial Ethernet and standards specific to the aviation industry. Additionally, the advent of low cost mobile telephone technology and wireless interconnect for PC and electrical goods is driving interest in the adoption of this technology in other areas. Already products are appearing on the market which exploit this technology and many applications exist. This overview presents an introduction and overview of the current state of the technology as a result of studies performed by the author for Rolls-Royce into potential exploitation of these technologies in Rolls-Royce aircraft, marine and industrial systems. In particular, the application of Ethernet and wireless technologies for on-board systems, remote operational monitoring, testing and control will be considered.

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1. The rise of Ethernet as a networking standard

Ethernet has been around for many years and is predominant in office computer systems with an 80% share of the LAN market. It has many advantages such as fast start-up, a simple connection system, high availability, ease of system extension, good communications performance and switching technology is available for scalable expansion of systems. The technologies for company-wide communication are also well developed with the possibility to interface to a Wide Area Network (WAN), such as ISDN or the Internet. Driven by the very large computer market there is also security of investment through constant further development. Thus, Ethernet is here to stay with future developments likely to be in faster versions, i.e. Gigabit Ethernet, and wireless options.

It is thus no surprise that industry is looking to Ethernet for a low cost networking solution. Early attempts were hindered by the fact that in “Classic

Ethernet” bandwidth is shared and not dedicated. Even if two hubs are interconnected, there is still one collision domain, with all traffic shared. (A hub is simply a multi-port broadcast device. It takes whatever comes in any port and broadcasts it out all the other ports.) As network nodes are added or traffic increases, every node in the collision domain has a greater chance of slowing communication or having a collision. It was thus impossible to predict message delays. Additionally, there was no concept of prioritised data so there is no way to differentiate between high priority and low priority traffic. It was thus possible for non-essential traffic on the network (perhaps people backing-up their computers to the network server or printing a large document across the network) to slow or collide with essential control traffic.

For light traffic this is not a problem as collisions will occur rarely, however, if the load on the network increases (through increased traffic or due to a large number of nodes) the chances of collision increases. This could be catastrophic when considering Industrial or Aerospace applications. A consequence of this is that

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“Classic Ethernet” is non-deterministic making it impossible to certify.

There were then two major advances. The first was the introduction of Bridges which act as a “gatekeeper” allowing two collision domains to be separated. The key advance, however, was the introduction of switches in 1993. These combine the multi-connectivity of a hub with the selective routing of Ethernet packets of bridges. Most modern switches have buffers that allow them to store and forward the Ethernet packets that are sent to it. Modern switches also have plug-and-play capability. This means that they are capable of learning the unique addresses of devices attached to them (even if those devices are plugged into a hub which in turn is then attached to the switch) without any programming. By 1999 Ethernet switches with performances of Terabits/second became available at much reduced prices. It was thus feasible to give all users a dedicated port.

Although cheap Ethernet networking technology became available for the office environment it still needed to be tailored for industrial use. As a result Industrial Ethernet was developed to provide a fast, deterministic and cohesive Ethernet backbone for the harsh conditions on the factory floor with built-in redundancy and durability so that all devices remain “connected”. On top of this backbone users can run familiar software, e.g. ModBus and Controlnet, so the devices can communicate with one another. Industrial Ethernet is still, very much, a developing standard. At higher levels some control suppliers have developed solutions using proprietary code residing above the standard TCP/IP stack. An example of this approach is Hewlett Packard’s implementation that allows synchronisation across nodes to around 200 ns (www.hp.com). The adoption of proprietary standards, however, leads to interoperability problems locking a user into one specific supplier. In an attempt to standardise and provide interoperability specifications are being developed by the Industrial Ethernet Association.

Many products are available and one of the key ways to differentiate between suppliers is to look at the Quality of Service (QoS) that their system promises. QoS is the speed, consistency and reliability with which data gets to its destination via the networks switches and routers. The Industrial Ethernet must be able to differentiate real-time applications from the rest and guarantee real-time performance over the network. This means that it must accommodate multiple service quality levels to ensure that mission-critical applications are not impacted by lower-priority traffic. Additionally, the technology must be stable, give high availability and integrate with the Office Automation (OA) and the enterprise networks. It is also likely that the system needs to be integrated with legacy systems and networks. Security, network management and long distance communications may also be key parameters.

Other developments are aimed at trying to improve throughput and fault tolerance. Fast Ethernet (IEEE specification 802.3u) is standard Ethernet operating at 100 Mbps or 200 Mbps in Full-Duplex mode (IEEE802.3x), ten times faster than standard 10 Mbps Ethernet. More recently, auto-sensing 10/100 Ethernet is becoming the standard Ethernet interface for users. For future scalability, when the desktop evolves to 100 Mbps, Gigabit Ethernet (IEEE specification 802.3z) will provide the bandwidth to remove bottlenecks to servers and between Ethernet switches. IEEE 802.12 provides the ability to add redundant links to the network to facilitate automatic recovery of network connectivity when there is a link or repeater failure anywhere in the network path.

2. The networked aircraft

On an aircraft the avionics systems represent a significant proportion of cost. The use of Commercial Off-The-Shelf Technologies (COTS) to reduce cost is thus very attractive and the adoption of PC-based hardware, router technology and Ethernet standards in place of the more traditional ARINC 429 and ARINC 629 standards offers many advantages. The Next Generation Aircraft will thus have a host of systems interconnected via a databus network as shown in Fig. 1. There are two main networks separated by a firewall. The Aircraft Systems Intranet connects avionics modules and the aircraft cockpit and reporting systems. Initially, interface units will be used to allow conversion of ARINC 429 databus protocols commonly used on aircraft to the Ethernet protocol so that a wide range of legacy hardware can be interfaced into the aircraft network. Additionally, future aircraft are likely to use a local wireless communication network to allow communication between the cockpit, the cabin crew and maintenance crews working around the aircraft. The second Cabin Intranet is driven by increasing passenger demands for entertainment (overhead screens and back

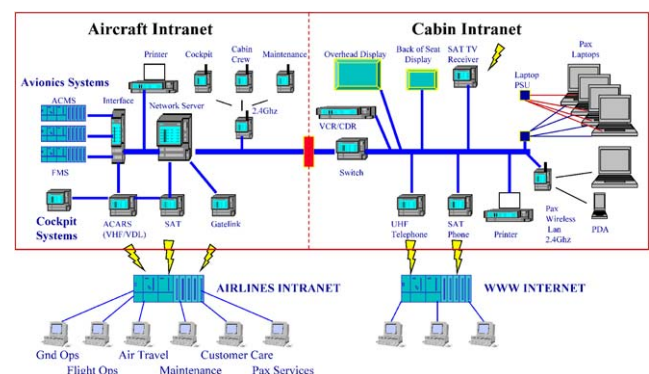


Fig. 1. The networked aircraft.

of seat displays) and connectivity to the World Wide Web using TCP/IP protocols.

Boeing and Honeywell are promoting the developing ARINC 664 standard (ARINC 2000, 2001) for aerospace with the intention of using COTS Ethernet technology, e.g. bridges, routers, switches and hubs in an aircraft environment. (Note that Airbus is promoting their own standard AFDX for the A380 which is also covered by ARINC 664.) ARINC 664 deals exclusively with Internet and IEEE standards for applicability to commercial networks in aviation. This includes flight deck systems and OEM systems where determinism is a key factor, Airline systems where the use of COTS is advocated (if possible) and passenger systems which are fully COTS. Aircraft Systems are defined in terms of Compliant or Profiled Aircraft Data Networks. ARINC 664 supports both. A Compliant Aircraft Data Network is a typical Internet-based system with the intention of providing maximum interoperability with ground-based Internet networks. For instance, it may provide data communications for passengers, e.g. e-mail and web browsing. Compliant Aircraft Data Network designs are expected to be the same as those found in desktop computers for use in the home or office. The integrator is free to implement whatever is best for the customer within the regulations RTCA DO-178 (RTCA, 2000). Development and certification methodologies are up to the designer. Commercial software can be used but the software needs to be written to RTCA DO-178B Level D or lower. Data transmissions may also require extra security, e.g. for financial transactions by passengers or protection of private messages. The Compliant Aircraft Data Network may also carry UDP/IP or TCP/IP routing and transport protocols above the Ethernet Data Link Layer.

A Profiled Aircraft Data Network is to be used when the system needs to be certified. In Internet no a priori knowledge is needed of the actual IP address. The application knows the name of the target but does not know the logical host which contains that target application. This allows independence of applications from the physical (or logical) architectural implementation. This lack of definition would present certification problems for an aircraft. Hence, in a Profiled Network the Address Resolution Protocol (ARP) table is statically defined at start-up (rather than through address discovery) to enable certification (Table 1).

Physically of the 13 Ethernet standards available ARINC664 (ARINC, 2001) recommends four:

- 10BASE-2 Coaxial, half-duplex only,
- 10BASE-T 2 twisted pair or 1star quad, half or full-duplex,
- 100BASE-TX 2 twisted pair or 1 star quad, half or full-duplex,
- 100BASE-FX Fibre Optic, full-duplex only,

although the faster versions are preferred for future expansion.

For the cabin systems, e.g. in-flight entertainment, file server, water and waste systems, the best commercial connectors can be used. For the avionics bay ARINC 600 style connectors should be used. (ARINC 600 defines sizes, mate-ability, contacts, connector blocks, racks, insertion forces and terminations to be used in rack mounted electronics.) For extreme environments, e.g. engine and wing installations, MIL-STD-38999 connectors should be used. For severe environments MILSTD38999 connectors are needed at both ends of the link with fewer than eight production breaks. (MIL-STD-38999 circular connectors should be used for production breaks.) It should be noted that the use of fibre optics and Ethernet connections in severe environments, i.e. on-engine, is still an area for research.

3. Remote health monitoring

Rolls-Royce have around 50 000 engines flying around the world. Additionally, they have engines in ships and land-based applications. High availability is a crucial operator requirement in both traditional contracts and new Total Care Packages where customers make fixed regular payments based on hours flown to cover engine maintenance. If the guaranteed availability levels are not achieved penalty costs must be paid.

These increasing reliability demands from operators are thus being met by the development of sophisticated on-board diagnostics and health monitoring. Consequently vast amounts of data (performance, event reports, prognostic information, trending and usage data) needs to be recorded during a flight/voyage. Ideally, this should be downloaded to local ground stations and passed via the internet to maintenance centres.

The initial hurdle is getting the data (typically a gigabyte) from an aircraft/ship or industrial engine to a suitable ground station. This needs to be done without huge infrastructure investment, e.g. at airports, so there is great interest in using existing telecommunications technologies and networks.

4. Telecommunications technologies

GSM (Global System for Mobile telecommunications) (Mouly & Pautet, 1992; Rahnema, 1993; Balston, 1991) has been phenomenally successful. (In Western Europe over 50% of the population has a mobile phone. In the UK the 50% mark was passed in May 2000). The success of GSM has been largely due to the European Commission's success in defining a standard that is interoperable. Although America and Japan use different

Table 1
Summary of wireless technologies

	Speed/frequency	Comparison with 56 kbits/s Computer Modem	Range/coverage	Comments	Application
GSM	9.6 kbits/s Uplink 890–915 MHz Downlink 935–960 MHz	6 times slower	Good network infrastructure over land. 120 countries. (Not Japan or America)	Mainly voice but Short Messaging Service SMS messages up to 160 bytes	Aircraft—over land/ near airport Marine—near to coast or port Land—remote access, e.g. pipelines
WAP	9.6 kbits/s As per GSM	6 times slower	As GSM	Slow to log onto network (4 mins). Limited by 5 line screen and 12 key keypad input	As above
Satellite	2.4 kbits/s 4–6 GHz	23 times slower	(Almost) worldwide	Faster data rates possible but long lead times and huge investment Expensive \$2–12 per minute.	Aircraft—on major flightpaths. Inmarsat Marine at sea. Land—remote access
Generation 2.5 Systems GPRS	Combines GSM channels to get bandwidth. A 3+2 GPRS handset: 40 kbits/s from Network 27 kbits/s to network	Depending on grouping of channels equivalent or up to 2 times faster	Exploits existing GSM infrastructure	Systems connected all the time. No log on time but scope for hacking/virus attack	Adds location capability. Can track resources over land. Useful for the automotive industry
Generation 3 Systems (UMTS)	560–1736 kbits/s 885–2025 MHz 2110–2200 MHz sub-bands 1980–2010 MHz/ 2170–2200 MHz for satellite	10–31 times faster	UMTS is based on GSM but needs huge investment in infrastructure	Systems connected all the time. No log on time but scope for hacking/virus attack. Pricing policies—flat rate or on bytes of data transmitted	Opens up new opportunities, location services, video, etc.
Wireless Ethernet	50–100 kbits/s at 900 MHz, 1–2 Mbit/s at 2.4 GHz (ISM) frequency. (5.8 GHz being introduced)	900 Mhz Equivalent 2.4 GHz 18–36 times faster	With 900 MHz 35 miles With ISM 300 m in open, 90 m inside	Cheap \$75 a node. Interoperability being Tackled. Hacking possible but at 2.4 GHz need to be local (300 m or line of sight with directional antenna)	900 MHz medium range 2.4 GHz Local monitoring and control
Bluetooth	1 Mbit/s at 2.4 GHz Industrial Scientific and Medical (ISM) frequency	18 times faster	10 m (works through walls)	Cheap \$3. Some Interoperability problems need resolving—UnPlugFests. Hacking possible but need to be local	Local monitoring and control

standards (TDMA—Time Division Multiple Access and CDMA—Code Division Multiple Access) triband phones are available for roaming worldwide, e.g. Nokia's "World Phone" works in 120 countries on all five continents. Services with GSM are, however, limited to telephony, data transfer, fax and Short Message Service. Provision of internet services is not practical with GSM because the bandwidth at 9.6 kbits/s is about 6 times slower than a computer modem. Off-the-shelf GSM interface chips are available for incorporating into equipment. ConnectOne Ltd. of Santa Clara (Grossman, 2000) produce the iChip Internet controller that provides an embedded Internet connectivity for devices via GSM modems at \$20. The most obvious application would be land-based industrial gas turbine engines, although it would also be possible to consider aircraft operating near airports. (Marine applications tend not to be in range of a cell.) There are, however, issues. Already some problems have been experienced by aircraft through GSM phone interference. SAS Commuter had to ground its Dash8 Q400 fleet after several instances of false smoke warnings from the cargo hold resulted in emergency landings. It is thought that the smoke detectors were being triggered by mobile phones in the luggage receiving incoming calls. As a result new smoke alarms had to be retrofitted to the fleet. This appears to be lack of foresight in design and it should be noted that wireless systems are used on board 90% of Boeing's business jets for fire detection and extinguishing. Even so, for aircraft, it would probably be advisable to disable any data upload or download during takeoff and landing.

Wireless Application Protocol (WAP) is based on GSM technology and has largely failed due to mis-selling of the technology as bringing internet to your phone. It has not lived up to its promise because of poor security, frustratingly slow speeds (especially to connect to a system—around 4 min), a small screen and cramped keypad. The display is typically five lines of text which makes display of a web page extremely difficult. Many companies have had to write web pages especially for WAP so that users can access information in a user-friendly way. In addition, the standard 12 key alphanumeric keypad of a mobile phone means that multiple presses are needed to get the 50 keys of a normal keyboard. This makes requesting information very time consuming. This can be alleviated by setting up standard replies from templates but again this requires effort on the part of the user. Even so companies such as Schneider have added WAP phone access to its internet enabled Programmable Logic Controllers (PLC) (Drives and Controls, 2000). This allows control and monitoring of the process from a mobile phone. In the system performance and diagnostic data can be downloaded using a web browser on request or the control system can raise an alert in the event of a system fault or at

predefined regular intervals. This can be done by sending an email or by using the mobile phone Short Messaging Service system which allows up to 160 characters. For security the WAP gateway asks for user identification (username and password) before connecting an engineer to the web server in the PLC. This is important as in WAP there is a separate wireless security layer. Unfortunately this is not the same as the one used on the public network and as the data is translated from one to the other it is exposed unencrypted for a moment at the gateway. Although this can only be exploited by the most determined hacker security advisers recommend that a company installs and operates its own WAP gateway to prevent a hacker getting access to the system.

Security is in fact a big issue when considering wireless communications. In Spain the "virus" Timofonica sent GSM mobile phone users messages. This used Microsoft Outlook to find peoples mobile phone numbers and send then a short Messaging Service Message. In Japan an "i-mode virus" caused Japanese mobile phones to produce a message inviting users to look at a website by pressing a key. This resulted in a phone call to the emergency services. In each case these were not a true "virus" as they could not replicate themselves or infect other phones. Indeed, currently a mobile phone virus does not exist but they are likely in the future. One reason for the lack of virus problems is the lack of functionality in a mobile phone to support a virus. This is, however, changing and if the ability to send scripts to other phones and automatic initiation of functions are added there are likely to be problems. Unfortunately, due to the pressure of users requiring more functionality this is just what companies like Motorola are doing, e.g. running Java applications. This is a concern as Microsofts Active X (which is similar to java) has been used by hackers to destroy the contents on a personal computer. In an industrial application there is thus a case for limiting the functionality of a system as a way of avoiding problems. As the connection is only made for data transfer additional security can be added by uploading and downloading data at unpredictable times.

Satellite communications make the most sense for maritime applications where satellite phones are commonplace. For aerospace applications the use of satellite communications has the promise of worldwide coverage; however, they are expensive to install and use. The bandwidth available (2.4 kbits/s) also restricts the amount of data that can be sent. Faster data rates are being introduced with new satellites but the long lead times for satellite systems means that they will always be behind terrestrial networks in terms of bandwidth and services. The main operators, e.g. Inmarsat and Globalstar, freely acknowledge that they cannot compete and are a niche market (Taverna,

2000). The main target market for satellite communications is business jets with calls costing typically between \$2 and \$12. For widespread use the market is tough and competition from terrestrial systems is fierce resulting in some notable failures, e.g. Iridium. Call costs will not reduce unless more passengers use satellite phones but there has been little success getting passengers to use on-board phones due to high cost. A new approach by Aerospatiale Matra Airbus is the In-Flight Telecom System (IFTS) (Paraco & Taverna, 2000) that will allow passengers to use their own personal mobile phones. Again this service is likely to cost \$2 per minute.

Before the introduction of generation 3 systems the introduction of generation 2.5 GPRS (General Packet Radio Service) systems offers more bandwidth than is available with a conventional computer modem (by using multiple GSM channels) opening up opportunities for internet access and a host of new services. Of interest are location services which are possible because the system is connected all the time. There is great interest here in the automotive industry where fleets of cars or lorries can be tracked to gather performance information on operation in the field.

Generation 3 systems bring 10–31 times the bandwidth of a computer modem opening up opportunities for more services and functionality from “always connected” systems with high data transfer rates, e.g. video clips beamed to airports may well be useful for maintenance engineers as well as the ability to take digital photos and send them back to base for diagnosis. Although fundamentally based on GSM the Universal Mobile Telecommunication System (UMTS) is being developed as a global standard. The global standardisation process is well developed which bodes well for universal adoption. The issue here is one of huge infrastructure investment which is difficult in the current telecommunications climate. NTT DoCoMo have been extremely successful in Japan with their i-mode system which allows music and games to be downloaded to a phone. They are also well ahead on 3G technology and will be introducing 3G systems into Japan ahead of Europe.

Although GPRS and Generation 3 systems open up many opportunities for development of multifunction devices and services only time can tell what “winning” functionality will be (the explosion of text messaging is a classic example of unpredicted success).

Security is an issue as GPRS and Generation 3 systems will be “on all the time” and hence the first real technologies that are susceptible to attack driving significant activity in security software development.

From a commercial standpoint the pricing policy adopted is important as these will depend on the data transferred. Two options are likely; charge per Mbytes of data sent or pay a flat rate. It should be noted that flat

rate charging may or may not succeed as mostly users will want to have high bandwidth for milliseconds at a time, interspersed with periods when much less capacity (or none at all) is needed. Paying a flat rate would allow a company to gradually increase the amount of data they transfer as the number of assets being monitored and system functionality is increased.

5. Distributed aircraft maintenance environment—dame

Once the data has been transmitted to the ground it needs to be collated at a centralised maintenance facility and acted upon. This requires high bandwidth to transfer thousands of gigabytes of information quickly, large computational resources to process it and provide diagnostic information in a timely fashion (within minutes for maintenance crews and hours for fleet management) and also massive storage capability for retaining data. Here the Sheffield UTC is working with the Universities of Leeds, York and Oxford, Rolls-Royce and Data Systems and Solutions on the Distributed Aircraft Maintenance Environment. This project funded by EPSRC's E-Science initiative is investigating the use of GRID technologies for dealing with maintenance data and providing decision support for aircraft maintenance.

6. Testing and development

For testing and development (and also possibly local data download to a PDA or portable computer) short-range communications are required. Of interest here is the globally available industrial, scientific and medical (ISM) unlicensed radio band operating at 2.4 GHz. There are two main technologies available Wireless Ethernet and Bluetooth. In the following sections these are discussed.

7. Wireless Ethernet

In home and office computing there is currently an explosion of cheap Wireless Ethernet technologies operating in the 2.4 GHz band (typically \$75 for a PC card with a range of 300 m in the open and 90 m in an office) with data rates of around 1–2 Mbits/s (note that conventional Ethernet operates at 100 Mbits/s). Typically, Wireless Ethernet offers multirate devices with speeds of 22, 11, 5.52 and 1 Mbits/s but the actual speed that is used is dependent on the receive sensitivity. At lower bit rates the receive sensitivity improves so that at 11 Mbits/s the receive sensitivity may only be –76 dBm but at 1 Mbit/s the receive sensitivity might

be—88 dB m. (Effectively better receive sensitivity means longer range.)

As Wireless Ethernet tends to be connected to a wired Ethernet it must conform to the 802.3 standard. This means that the systems are designed to match to 10 Mbits/s network speeds. The emphasis is on speed and not range whereas in an industrial environment the emphasis tends to be on range rather than speed. Two types of spread spectrum systems are employed, direct sequence and frequency hopping. For industrial applications there is some evidence that frequency hopping is advantageous particularly as with 64 hopping.

Frequencies up to 16 co-located networks can be supported. (With a direct sequence system only three separate links can be accommodated in the same location.) Originally, IEEE 802.3 was used to cover Wireless Ethernet but this did not incorporate encryption (it was assumed that it would be used in buildings with controlled access). However, to ensure privacy and also tackle compatibility issues the Wireless Ethernet Compatibility Alliance (WECA) was formed. WECA have introduced IEEE 802.11 High Rate which specifies wireless Ethernet quality, interoperability and encryption. All equipment that meets this standard can use the Wi-Fi logo. It must be noted that even though a 128bit Wired Equivalent Privacy Encryptor can be used (each packet is encrypted and the user has the key) it is still advisable to add additional security, e.g. SSL. This is because it is possible to get into these systems although with 2.4 GHz systems the attack must be local (due to the limited range of the system) and a compatible wireless card is needed (not a radio scanner). Even using a directional antenna the attacker still needs to be operating line of sight.

In parallel with developments in home and office wireless Ethernet there is also an increasing number of equipment providers for the industrial environment, e.g. Data-Linc, BenteK, Siemens, Synetcom, Cirronet, Microhard and Unicom (Maes, Mhoon, Currie, & Ralston, 1999).

Although wireless Ethernet is of interest in aerospace there is considerably more interest in Bluetooth as this allows low cost, single chip solutions. The remaining sections will consider the application of Bluetooth in a number of areas.

8. Personal area networks for testing and development

The basic Bluetooth technology (Vollmer, 2000) is an inexpensive (\$3), low power, short-range radio on a chip that can be embedded in any device to translate digital data from computers. It is finding wide application in computers, PDAs, phones, cameras, etc. with enormous market predictions (Vollmer, 2000; Varshney,

2000). There are over 2000 companies working on applications. The radio sends and receives voice and data signals generated by other Bluetooth radios that come within the broadcast range (10 m). Since radio waves pass through solid substances Bluetooth devices can communicate through walls and other barriers that stop competing technologies such as infrared. (Bluetooth devices also do not require line of sight to communicate.) The antenna for Bluetooth products is a short segment of conducting track on a printed circuit card. The radio operates on the globally available industrial, scientific and medical (ISM) unlicensed radio band with frequencies between 2.402 and 2.408 GHz using frequency hopping with 1600 frequency changes per second. A single Bluetooth connection uses 79 different frequencies with a channel separation of 1 MHz. Three voice channels are supported as well as data transfer. Each data packet is transmitted at a different hopping frequency. (Frequency hopping is used to provide noise immunity as packets corrupted at one frequency can be re-transmitted at another. Note although this is also claimed to give added security the hops are predictable and it is possible to buy equipment that can synchronise to this.) Non-voice data can be transmitted either asynchronously at 721 kbits/s upstream and 57.6 kbits/s downstream or synchronously at 432.6 kbits/s in both directions. With control and protocol bits this is equivalent to 1 Mbits/s.

The connection itself is controlled by the Bluetooth link controller which is part of the baseband IC. This controls the protocol and link access routines. In the beginning all nodes in the piconet are in stand-by mode. Every 1.28 s they listen for a signal. If a signal is detected the Bluetooth module will look for a partner on 32 frequencies by sending telegrams. The maximum delay in contacting a slave is 2.56 s although in most cases only half of this time should be needed. An inquiry telegram is then used to identify Bluetooth devices with the new unknown address. If no data is transmitted the master can tell the slaves to go into a hold mode. This is also the case when the master wants to talk to another piconet using the scatternet. Within 2 s individual Bluetooth controllers within a piconet identify themselves using a unique 48 bit serial number. The first device identified becomes the master and controls the 1600 frequency hops per second. All users participating on the same piconet synchronise to this hopping sequence. Several piconets can communicate with one another over a “scatternet” with communication occurring between the piconet masters. Each of Bluetooth’s picos, or cells, can accommodate up to eight PANs within a 100 m² range. Bluetooth works with short data packets and so the full duplex data rate within a scatternet that has 10 fully loaded piconets is 6 Mbits/s.

9. Bluetooth for test bed monitoring

One potential use for Bluetooth wireless communications is in test bed monitoring of engines (see Fig. 2). Instrumenting an engine is a time consuming task requiring connection from an enclosed test cell to a protected control room. The ability to simply wire a sensor to a Bluetooth transmitter and then collect the signals via radio through the test cell walls would be a great advantage. Indeed there is some activity in the engine world investigating this for reducing test times by drastically shortening the time to instrument an engine to allow significantly more time for engine testing. Additionally, the number of sensors used in testing is significant (around 2000). Many of these are replicated so that if a sensor fails (normally due to a wiring fault) during an engine test the results are not negated.

By using a wireless approach many of these problems could disappear allowing far fewer sensors to be required. At the Sheffield UTC, Bluetooth modules from Crossbow Technologies (Alan, 2000) are being investigated to identify suitability for test cell monitoring.

10. Wiring reduction on aircraft

Wiring on an aircraft is complex, difficult to route, heavy and a key source of faults which lead to delays and cancellations for passengers. Engineers thus try and minimise wiring as “every wire is a source of failure and every wire adds weight”. An A340 for instance has 152 miles of wire weighing 1761 kg, of which 50% is for power distribution. The problems of open and short circuits, as well as intermittent faults from degraded contacts inside connectors, caused by vibration are well known. The aging of wiring is also becoming more of an issue in older aircraft with studies identifying cracks in insulation (one recent study found three insulation breaks per 300 m of wire taking a random sample of

three different aircraft types) and numerous intermittent faults attributed to changing conductance to ground from poor airframe ground connections which are particularly difficult to isolate.

Inside the fuselage in pressurised areas the environment is fairly benign. Here there are clear advantages to using wireless communications and research work on wireless systems is being performed at the Aircraft Vehicle Systems Institute in America and ARINC are working on specifications for wireless systems in aircraft. In fact the metal fuselage of older aircraft acts as a fairly good screen, however, the increasing use of composites in modern aircraft structures presents problems (Fig. 3).

Of interest to Rolls–Royce is on-wing monitoring. (Particularly for specialised “bolt-on” monitoring for flight testing). There are, however, a number of problems which need to be overcome. The environment is harsh in unpressurised areas and on the fan case of an engine temperatures can drop to -55°C with exposure to vibration, flexing, stress, solvents, lubricants and fuel. On the core of a gas turbine engine temperatures may be above 500°C requiring high temperature wire and connectors. Although there is an interest in using wireless systems it is difficult to find electronics that will survive in the environment. Some work is being performed in America flight testing Oceana Bluetooth sensors on a C17 aircraft recording pressure data at the front of the engine and also monitoring RF emissions.

Although this is a necessary first step in acceptance of the technology there are still a number of limitations which need to be considered. The engine sensors (see Fig. 4) can be split into high bandwidth (above 20 KHz for vibration monitoring), medium bandwidth (around 10 KHz for pressures and speed), low bandwidth (less

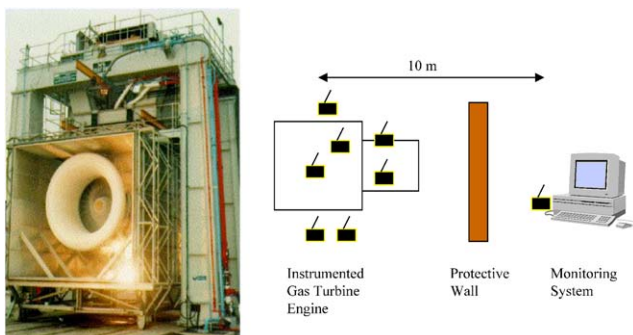


Fig. 2. Test bed monitoring.

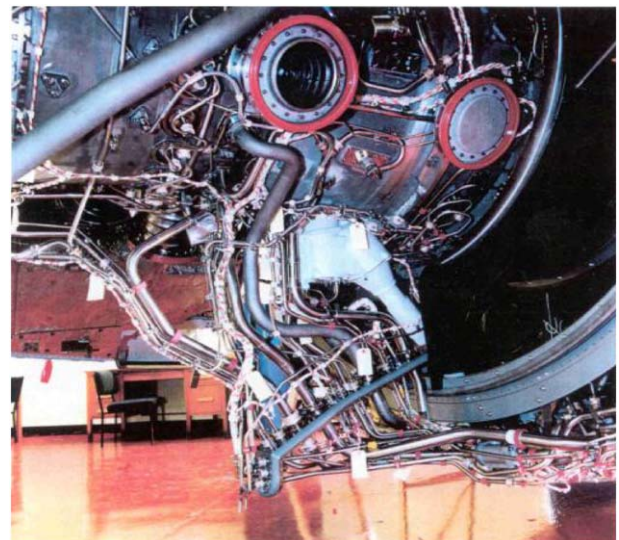


Fig. 3. Typical gas turbine engine wiring harness.

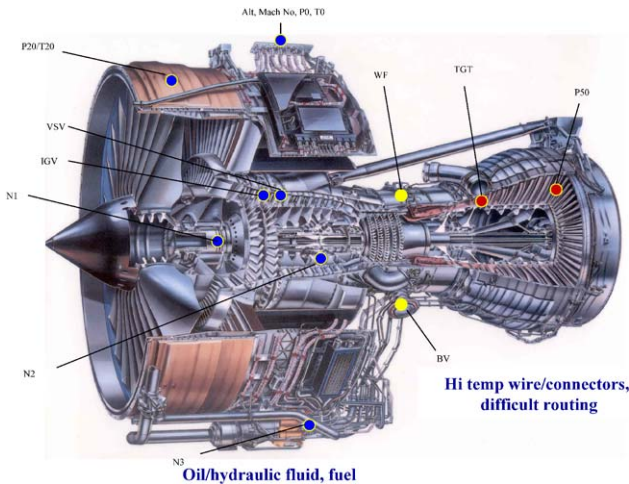


Fig. 4. Engine sensors.

than 1 KHz for temperatures, oil monitoring, fuel flow, fuel, oil and hydraulic pressure, Pilots lever, Inlet Guide Vanes and oil quantity) and discretes (valve position open/shut and thrust reverser interlock position).

Although the bandwidth of Bluetooth is nominally 1 Mbit/s it is important to consider what sample rates are achievable. Bluetooth is designed to operate with voice and data packets. There are six types of data packets that can be specified DM1, DH1, DM3, DH3, DM5 and DH5 transferring increasing amounts of data from 17 bytes per message to 339 bytes per message.

(The DH versions operate slightly faster due to the fact that some checking is switched off.) Considering a system composed of individual sensors producing 12 bit data each with their own RF interface transmitting DM1 packets at most a packet can be transmitted once every 1350 μ s and so around 800 packets can be sent a second.

The currently available commercial Bluetooth systems, can thus only sample in the millisecond range confining use to low bandwidth and discrete position monitoring.

Consideration of the protocol overheads if each sensor sends it's own packet is even more illuminating. For every 12 bits of data sent there would be an overhead of ACL data packet = 72 bit access code, 54 bit header, 16 bit CRC + 17 bytes data (for DM1) giving 278 bits. This equates to 96% of the bandwidth being lost due to the communication protocol. This immediately highlights the fact that in a practical system sensors would need to be grouped together with a Bluetooth interface. Even with eight sensors wired into a single Bluetooth transmitter the protocol overhead is still 65%. To get the best out of the system careful optimisation of data flow is necessary.

11. Providing power

There is no real advantage to removing signal wiring unless the need for power wiring is also removed. When transmitting a Bluetooth device should consume 8–30 mA (or less than 1/10th of a watt) and in “hold” mode around 30 μ A. To conserve power Bluetooth also has sniff and park modes. In sniff mode they can scan at a lower rate dictated by the application. In park mode (where the least power is used) the slave module is synchronised within the network but does not transmit unless a wake up call is sent. Typically, Bluetooth modules only require about 2 mW in the power saving modes making them suitable for battery operation. Standalone sensors could use batteries but for a fit and forget, low maintenance system work on energy harvesting is needed to convert heat and vibration energy into power for the sensors (studies at MIT have shown that around 20 mW is available on a gas turbine engine).

12. Wireless aircraft monitoring network

One interesting application for a wireless network would be aircraft health monitoring, e.g. structural monitoring of an aircraft using strain gauges. Fig. 5 shows a Boeing 777 aircraft and Tornado aircraft to scale considering coverage of Bluetooth cells. For the commercial aircraft with suitable location of only 10 Bluetooth transmitters it would be possible to provide a wireless communications network that would cover the entire aircraft. This would need to exploit both the piconet and scatternet functionality of a Bluetooth network. A military aircraft would be covered by a single cell. Recorded data could be downloaded post-flight via a PC or PDA as ground staff walked around the aircraft and of course this in turn could be connected to the Internet.

13. Bluetooth in control systems

Companies such as ImpulseSoft, Con1, connectBlue and CSR all provide Bluetooth equipment for industrial

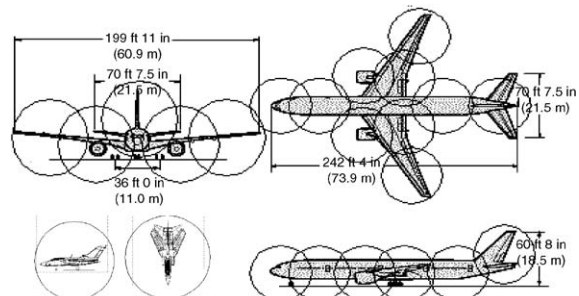


Fig. 5. Bluetooth monitoring network.

applications. The products, however, are targeted towards providing Bluetooth access points onto other networks such as Ethernet, Controlnet or Profibus. Some work has been done by ABB in Norway (Kjesbu & Endresen, 2000) investigating Bluetooth performance in a chemical pulp factory and there is considerable work going on investigating the use of Bluetooth in the US Navy (surprisingly Bluetooth signals propagate through the rubber seals around watertight doors) (Daniel, Estes, Sarkady, Welch & Whitese, 2003; Hogan, 2003; Harman & Clark, 2003). There are many advantages to using wireless systems on ships to remove cabling and also the need to pass cables through bulkheads which weakens the ships structure. Applications include diagnostics, rapid battle damage assessment and control of valves for firefighting. For safety-critical control certification becomes a big issue. For control of a passenger aircraft it is highly unlikely that a wireless system would ever be certified. Already there is concern about the use of mobile phones and their impact on aircraft instrumentation. If passengers had Bluetooth enabled equipment they could become part of the network. A passenger may well interfere to such an extent that control is interrupted or lost. Microwave ovens can also interfere with Bluetooth if very close. Another key problem at present is that Bluetooth is non-deterministic although there are plans to rectify this in future versions with the introduction of scheduled transparent data packets (which are sent according to a strict schedule even if they contain faults).

One interesting area which could be investigated is to use a wireless network in place of a backup lane of wiring to provide a redundant channel in case of failure. This would have to be carefully considered for certification. There is more scope for application in military aircraft and unmanned aerial vehicles where wireless backup lanes and control channels may be more acceptable. Wireless stores delivery has advantages although again there are great concerns about jamming or unintentional release through interference. Additionally, for military aircraft RF emissions are an issue from the point of view of stealth although the emissions from Bluetooth are very small and localised.

14. Concluding remarks

It is clear that industry will be driven by developments in mass market PC networking technology and wireless systems. Industrial engineers will use or adapt off-the-shelf technologies but will not drive development of new wireless technologies. This paper has attempted to provide an overview of possible applications in the aerospace industry using COTS hardware although parallels exist across other domains. Future aircraft will become local intranets and effectively an aircraft will

become a website with pages for each of the aircraft systems downloading and uploading information from ground maintenance and fleet management facilities. It is highlighted that the technology is already here to allow maintenance crews and test engineers to exploit wireless systems in test beds, remote monitoring and maintenance download. Current systems are limited by bandwidth but with developments this barrier will be overcome.

The next critical barrier is security and here much work needs to be done before wireless systems become accepted for transmission of commercially sensitive data (i.e. performance monitoring data). It is likely that initial applications will be in monitoring on test beds, followed by perhaps “one-off” special monitoring systems for in-flight testing, leading to wider application for in-flight monitoring. Once established, providing wireless backup fault tolerance channels and control systems may be considered although here there are significant certification hurdles to overcome.

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